

Arjun Rao

Machine Learning / AI Engineer

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Professional Summary

Machine Learning Engineer with 4+ years of experience building production-grade ML systems, specializing in deep learning, NLP, and computer vision. Proven track record of improving model accuracy, reducing inference latency, and deploying scalable ML services. Strong software engineering background with end-to-end ownership from data collection to deployment.

Key Skills

- Languages: Python, SQL, Bash
- ML Frameworks: PyTorch, TensorFlow, scikit-learn
- NLP / CV: Transformers, Hugging Face, OpenCV
- Tools: Docker, Kubernetes, MLflow, Airflow, AWS (S3, EC2, SageMaker)
- Data: Pandas, NumPy, PostgreSQL, BigQuery
- Deployment: FastAPI, Flask, Flask-RESTful, gRPC
- Other: Git, CI/CD, Unit Testing, Experiment Tracking

Professional Experience

Senior Machine Learning Engineer — Helios AI (Bengaluru) (Jan 2023 – Present)

- Led design and deployment of a recommendation system that increased user engagement by 18% using hybrid recall+ranking architecture.
- Built and productionized a multi-task transformer model (NLP) for intent detection + entity extraction, improving F1 by 12% over baseline.
- Reduced model inference latency by 3x through model quantization and TorchScript; integrated into Kubernetes serving with autoscaling.
- Implemented CI/CD for model retraining and deployment using Airflow & MLflow; enabled daily model refreshes with automated testing.

Machine Learning Engineer — DataForge Labs (Aug 2020 – Dec 2022)

- Developed computer vision pipelines for defect detection using Faster R-CNN and YOLOv5 achieving 91% mAP on production dataset.
- Built ETL pipelines and feature stores for time-series forecasting models (sales forecasting) resulting in 8% forecast error reduction.
- Collaborated with data engineering to migrate training workloads to AWS SageMaker, reducing training time by 40%.

AI/ML Intern — Cortex Research (Jun 2019 – Jul 2020)

- Implemented transfer learning experiments for text classification using BERT; improved accuracy by 7% on labeled dataset.
- Built automated data labeling scripts and augmented training data via back-translation and data augmentation for text and images.

Selected Projects

Semantic Search Engine (Prod)

Designed a semantic search pipeline using SBERT embeddings and FAISS indexing; achieved 92% retrieval recall and sub-100ms query latency.

Document Understanding (NLP)

End-to-end system for document parsing, entity extraction and QA using Hugging Face transformers and custom rule-based post-processing.

Vision-based Quality Inspection

Built a production CV pipeline using YOLOv5 + custom augmentation; integrated with automated alerting and dashboarding.

Resume generated for evaluation / example purposes — fictional candidate.

Education & Certifications

- M.Tech. in Artificial Intelligence, Indian Institute of Technology (IIT) – 2019